

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -4



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Recap of Previous Lecture



Doubt ?



Topic

HRRN Scheduling

Topic

Priority Based Scheduling

Topic

Round Robin Scheduling

Topics to be Covered



Topic

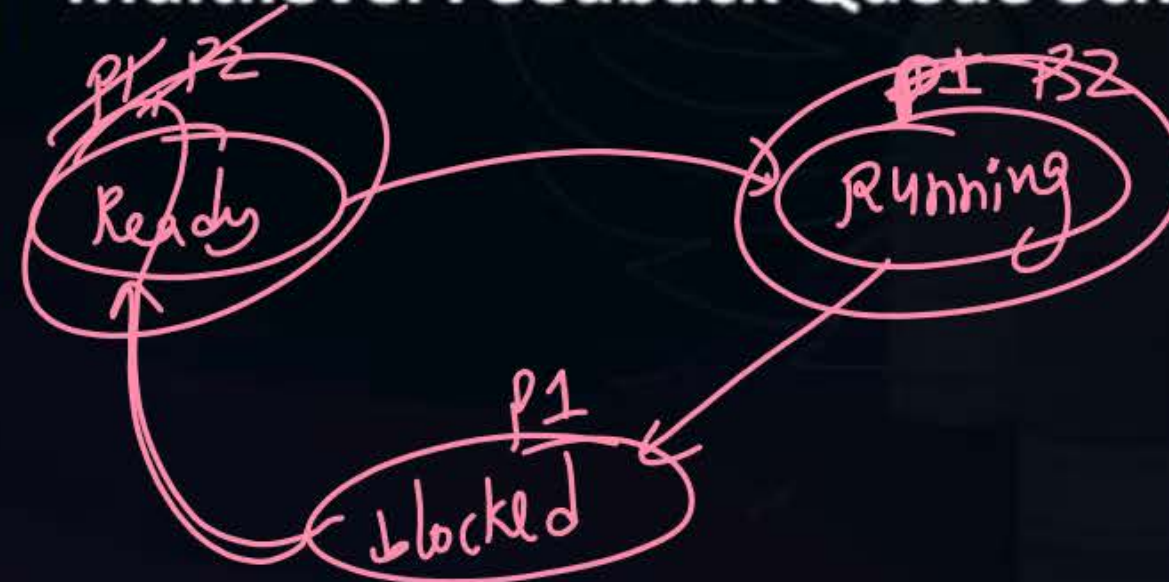
Round Robin Scheduling

Topic

Multilevel Queue Scheduling

Topic

Multilevel Feedback Queue Scheduling





Topic : Round Robin (RR)

$Q = 3$

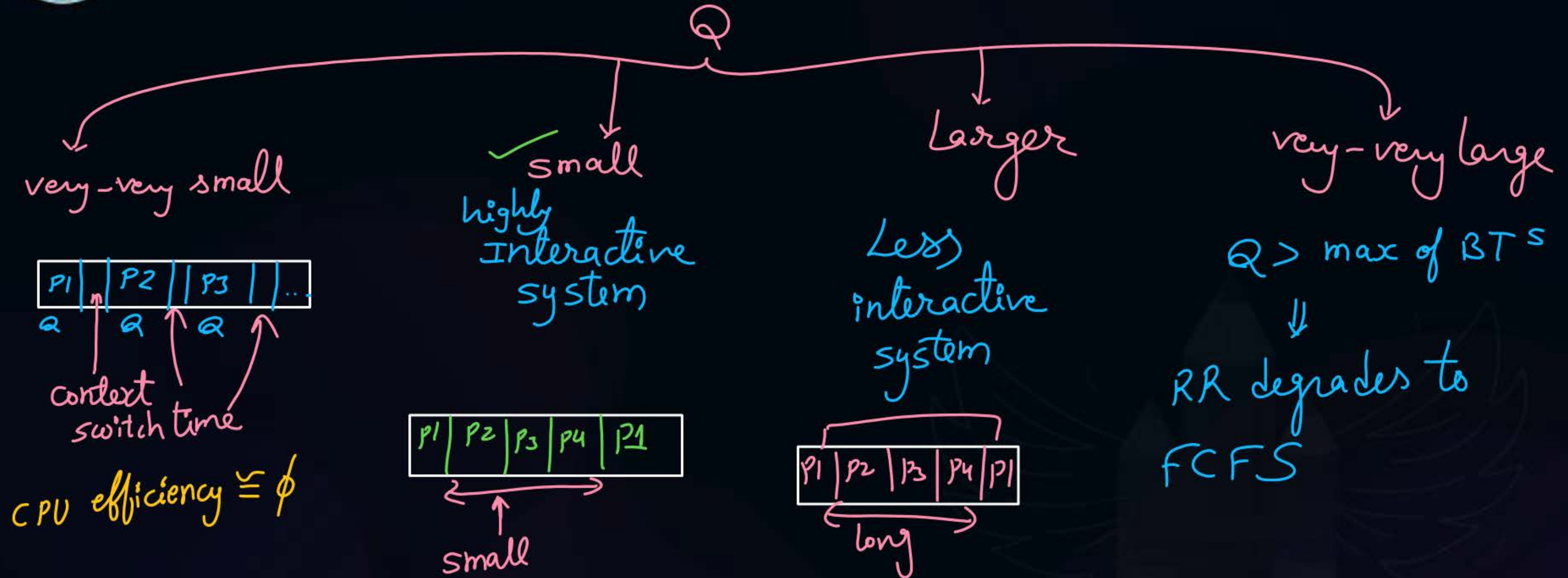
Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7

$$\text{avg TAT} = 35.57$$
$$\text{avg WT} = 28.57$$





Topic : What Should Be the Quantum Value?



$$\text{CPU utilization or efficiency} = \frac{Q}{Q + \text{context switch time}}$$



Topic : RR



Advantages:

1. All processes execute one by one, so no starvation
2. Better interactivensess
3. Burst time is not required to be known in advance $\Rightarrow$ hence practicall implemented

Disadvantages:

1. Average waiting time and turnaround time are more
2. Can degrade to FCFS

#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

GATE-2013

- A** This algorithm is equivalent to the first-come-first-serve algorithm
- B** This algorithm is equivalent to the round-robin algorithm.
- C** This algorithm is equivalent to the shortest-job-first algorithm
- D** This algorithm is equivalent to the shortest-remaining-time-first algorithm



Topic : Multilevel Queue Scheduling

1. have multiple ready queue.
2. keep processes in different queues as per some criteria

Ready Queue 1

⇒ algo 1

R.Q. 2

⇒ algo 2

R.Q. 3

⇒ algo 3

3. have different algo for each queue





Topic : Multilevel Queue Scheduling

a way to distribute processes in multiple queues is, basis on their type

System Processes

Foreground Processes

Background Processes





Topic : Multilevel Queue Scheduling

when to run processes of which queue

1. Fixed priority preemptive scheduling method $\Rightarrow$ Queues are give priorities.
 2. Run processes of higher priority queue first.
 3. when higher priority queue is empty then only lower —||— processes can execute.
 4. A low priority queue process can be preempted if any process arrives in high priority queue.
2. Time slicing
 - $\Downarrow$

50% time for Queue 1 process	30% time for Queue 2	20% time for Queue 3
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Topic : Multilevel Queue Scheduling

fixed priority preemptive

Queue 1: RR with Q=2

high priority

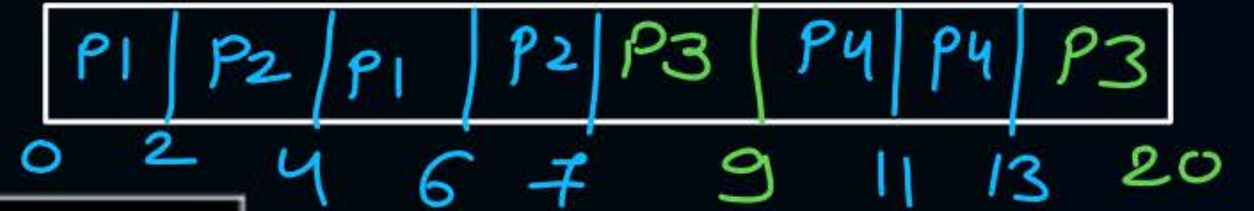
Queue 2: FCFS

(low)

At Time 0

Queue 1 : $\checkmark P_1, \checkmark P_2$ ~~$\checkmark P_4$~~

Queue 2 : P_3

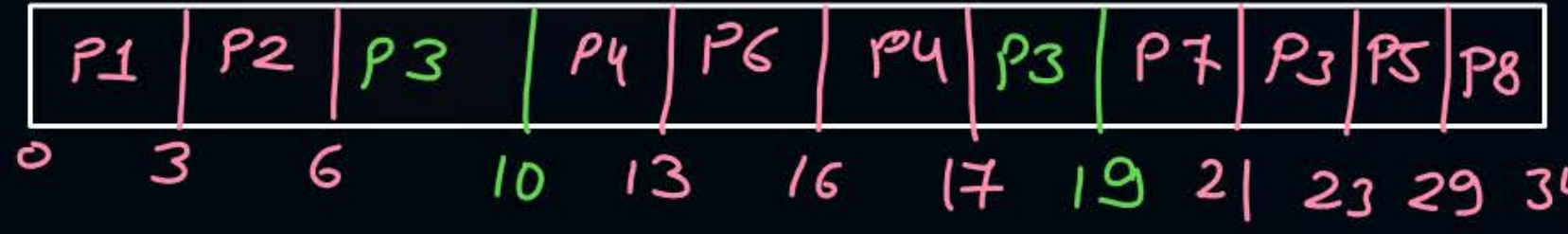


Process	Arrival Time	Burst Time	Queue
P1	0	4	1
P2	0	3	1
P3	0	9	2
P4	9	4	1



Topic : Multilevel Queue Scheduling

Queue 1: RR with $Q=3$ ~~P1, P2, P4, P6, P4, P7~~
(High)



Queue 2: FCFS P3, P5, P8

Process	Arrival Time	Burst Time	Queue
P1	0	3	1
P2	0	3	1
P3	2	8	2
P4	10	4	1
P5	11	6	2
P6	11	3	1
P7	19	2	1
P8	13	5	2



Topic : Multilevel Queue Scheduling

Disadvantages:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty
2. It is inflexible in nature. *⇒ The processes can not be moved from one queue to another queue.*



Topic : Multilevel Feedback Queue Scheduling

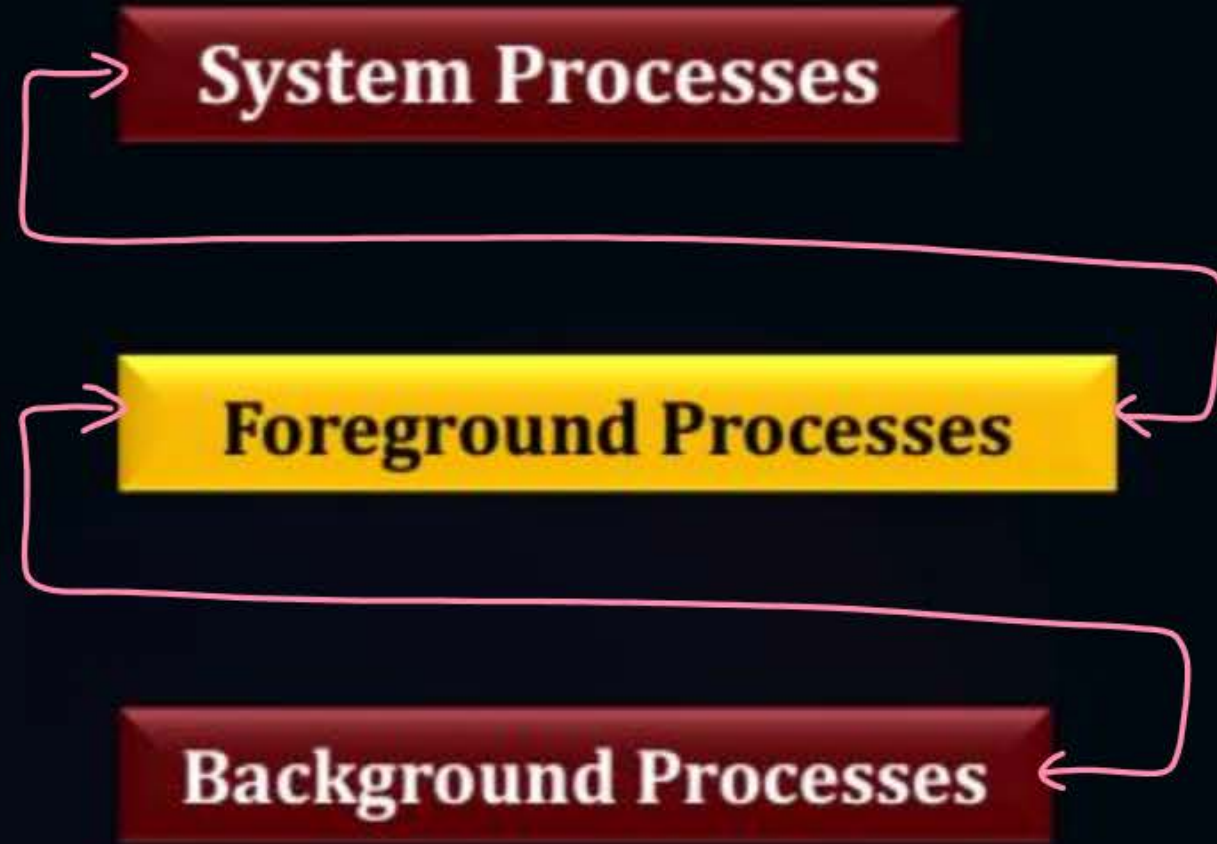
↓
Same as multilevel queue scheduling but processes

Can be moved from one queue to another queue.

→ Practically implemented as it includes advantages of RR and priority based algo^s.



Topic : Multilevel Feedback Queue Scheduling





Topic : Multilevel Feedback Queue Scheduling

Disadvantage:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty.

Advantage:

- ✓ 1. Flexible



Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
Minimum average WT among non-preemptive	SJF
Minimum average WT among all algo ^s	SRTF
Non-preemptive always	FCFS, SJF, LJF, HRRN, Non-preemptive priority
Preemptive always	None
SRTF behaves as SJF (no preemption in SRTF)	1. when all processes arrive together. 2. when later arriving process never has $BT < \text{Remaining time of current process}$
Preemptive Priority behaves as Non-Preemptive	1. when all processes arrive together 2. when later arriving process never has higher priority than current running process's priority.



Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
RR behaves as Non-Preemptive (FCFS)	when $Q > \max(BT^s)$
Convoy Effect	FCFS, LJF, LRTF
Starvation	SJF, SRTF, LJF, LRTF, Priority based algo ^s , MLQ, MLFQ

[MCQ]



#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses non preemptive SJF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time for the execution for the processes?

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time	CT	TAT	WT
P1	0	2	5	5	14	14	2
P2	0	4	5	2	16	16	5
P3	0	3	9	4	20	20	4
P4	0	6	4	1	31	31	26

$$\text{avg WT} = \frac{31}{4} = 7.75$$

TAT - Sum of CPU + I/O burst times



$$\left| \frac{P1 \text{ I/O}}{2+5=7} \right| \rightarrow \left| \frac{P2 \text{ I/O}}{9+5=14} \right|$$

$$\left| \frac{P4 \text{ I/O}}{26+4=30} \right|$$

$$\left| \frac{P3 \text{ I/O}}{5+9=14} \right|$$

Ready Queue: - ~~P1~~, ~~P2~~, ~~P3~~, ~~P4~~, ~~P1~~, ~~P2~~, ~~P3~~

#Q. Consider a process scenario in which each process executes first in CPU then goes for IO operation, then once again process needs a CPU bursts and then terminates. Following is given a process scenario in which for CPU execution system uses preemptive SRTF algorithm. Consider system has enough number of resources to carry out IO operations for all processes in parallel at a time. What is the average waiting time for the execution for the processes?

Process	Arrival Time	CPU Burst Time	IO Burst Time	CPU Burst Time
P1	0	6	7	1
P2	1	4	2	9
P3	2	1	6	5

- #Q. The arrival time, priority and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I.O burst followed by another CPU burst. Assume that each process has its own I/O resource.

Process	AT	Priority	BT (CPU)	BT(I/O)	BT(CPU)
P1	0	2	1	5	3
P2	2	3(lowest)	3	3	1
P3	3	1(highest)	2	3	1

The multi – programmed operating system uses preemptive priority scheduling. What are the finish times of the process P1, P2 and P3 ?

GATE - 2006



11, 15, 9



10, 15, 9



11, 16, 10



12, 17, 11



2 mins Summary

Topic

Round Robin Scheduling

Topic

Multilevel Queue Scheduling

Topic

Multilevel Feedback Queue Scheduling



Happy Learning

THANK - YOU

